

JSON Hijacking with UTF-7

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Content

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JSON Hijacking with UTF-7

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Attacking with Character Encoding for Profit and Fun POC2008 Yosuke HASEGAWA hasegawa@utf-8.jp

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Yosuke HASEGAWA NetAgent Co., Ltd R&D dept. Microsoft MVP award for Windows Security
Investigating about the security issues that a character code such as Unicode causes Discovered a lot of vulnerabilities including IE and Mozilla Firefox so far, such as CVE-2008-4020, CVE-2008-0416 , CVE-2008-1468, CVE-2007-2225, CVE-2007-2227 and more... Who am I? <http://utf-8.jp/>

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Introduction Comparison: match/unmatch Redundant encoding Many-to-one Conversion Upper case and Lower case Normalization Embedded invalid characters Embedded leading bytes Mismatch in charset information Interpreting 7-bit encoding Deceptive indications Characters with similar appearance Invisible characters Embedded control characters Conclusion Agenda はじめに 比較の一致 / 不一致 冗長なエンコーディング 多対一の変換 大文字と小文字 正規化 不正なバイト列の埋め込み 先行バイトの埋め込み エンコード情報の不一致 7 ビット文字コードの解釈 表示上の欺瞞 視覚的に似た文字 見えない文字 制御文字の埋め込み まとめ

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Introduction はじめに

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What is the relation between charsets and security? 文字コードとセキュリティ、何が関係あるの？

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Web browser is Text Parser Handles text data such as HTML/XML... Web ブラウザはテキストパーサ HTML や XML などのテキストデータを処理 ... What's the relation between charsets and security ?

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Upgrading from legacy encoding to Unicode. EUC-JP / Shift_JIS are often mixed in Unicode レガシーな文字コードから Unicode への移行 EUC-JP や Shift_JIS と、Unicode の混在 What's the relation between charsets and security ?

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Visual effect Similar lettes could be effective tools for attackers 視覚的な効果 視覚的に似た文字など、攻撃者の強力な道具 What's the relation between charsets and security ?

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Comparison: match/unmatch 比較の一致 / 不一致

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String comparison and detection Basic processing for security "confirm SAFE string to pass" or "detect DANGEROUS string" 文字列の比較検出 セキュリティのための基本処理 「安全な文字列の確認」や「危険な文字列の検出」 Comparison: match/unmatch

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Valid Invalid Overlong forms of UTF-8 One of the traditional attack techniques UTF-8 の非最小形式 伝統的な攻撃手法のひとつ Redundant encoding / U+002F 0x2F 0xC0 0xAF 0xF0 0x80 0x80 0xAF 0xE0 0x80 0xAF

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MS00-057 is famous. Currently, attacks like this have already become fossils.. IIS の MS00-057 が有名 もはや化石のような攻撃手法 Redundant encoding

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"fossils", Really? ほんとに化石？

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CVE-2008-2938 Apache Tomcat UTF-8 Directory Traversal Vulnerability Published: Aug 12 2008 Still existing issue, not past, "Living Fossil". いまでも存在する「生きた化石」 Redundant encoding

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Countermeasure: Don't implement functions handling UTF-8 yourself. Convert all strings into UTF-16 beforehand 自前で UTF-8 を扱わない 処理前に UTF-16 などに変換する Redundant encoding

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Conversions from Unicode to others has several "many-to-one" pairs. Unicode から他の文字コードへの変換は多対一で行われる Many-to-one Conversion U+005C ¥ U+00A5 \ 0x5C ¤ U+20A9 In Japan, path delimiter is displayed as YEN-SIGN.

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Many-to-one Conversion Input string as Unicode Validation Processing ¥..¥..¥ \ .. \ .. \ Bypass filtering Path traversal U+00A5 U+005C Convert to other encodings

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".." and "..\..\Windows" is existing in "C:\temp" folder. Path traversal occurs when handling filenames as ANSI. ファイル名を ANSI で扱うとパストラバーサル Many-to-one Conversion

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Many-to-one Conversion DEMO

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A lot of letters converted from Unicode are "many-to-one". Many-to-one Conversion U+00A1 U+00A6 U+00C0 U+00C1 U+00C2 U+00C3 À ; ¡ Á Â Ã U+00C4 U+00C5 Ä Å U+00C6 Æ 0x41 A 0xA5 0x7C ! | 多数の文字が多対一で変換

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Countermeasure: Handle strings as Unicode, without conversion. Don't convert after validation, even if conversion is necessary. Unicode のまま文字列を扱い、変換しない (変換するとしても) 検査後には変換しない Many-to-one Conversion

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Definition of the identification for Upper-Case and Lower-Case is different by a language culture. 大文字、小文字同一視の定義は、言語文化によって異なる Upper case and Lower case

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Comparison of Upper-Case and Lower-Case Upper case and Lower case Word 単語 Equivalent 一致 Nonequivalent 不一致 Gif / GIF U.S. アメリカ Turkey トルコ Ma ße / MASSE Germany ドイツ U.S. アメリカ Ma ße / Masse Switzerland スイス Germany ドイツ U.S. アメリカ 「Windows プログラミングの極意」, 株式会社アスキー, ISBN978-4-7561-5000-4, P.340 より

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Countermeasure: Don't adopt difference between lower case and upper case as boundary of security. Never rely on case-conversion rules you expect. 大文字、小文字の差でセキュリティ上の分界点をつくらない Upper case and Lower case

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Unicode supports the Composition and Decomposition of letters. No differences in appearance, but byte sequences are different Unicode は文字の分解・合成をサポート 見た目は同じでもバイト列が異なる表現 Normalization U+304C U+304B が か U+3099 〃 Precomposed character Base character Combining Character

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Unicode defines four specific forms of normalization. NFC Normalization Form Canonical Composition
NFD Normalization Form Canonical Decomposition NFKC Normalization Form Compatibility
Composition NFKD Normalization Form Compatibility Decomposition Cannot restore original byte
sequence after Normalization. Unicode では 4 種類の正規化方法を規定 正規化した結果から元のバイト
列の復元はできない Normalization

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Normalization process changes the byte sequence into another of different meaning 正規化により意味
の異なるバイト列に変化 Normalization U+2025 U+002E .. U+002E . U+2473 U+0031 ① 1 NFKC,NFKD

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Normalization Input string as Unicode Validation Normalization Processing ¥ .. ¥ .. ¥ \ .. \ .. \ Bypass
filtering Path traversal U+2025 U+002E

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Countermeasure: Never normalize strings after validation. 文字列の検査後に正規化を行わない
Normalization

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Depending on the implementation, illegal byte sequence is often ignored or converted to unexpected
characters. 処理系によっては不正なバイト列が無視されたり、想定外の文字に変換されることがある
Embedded invalid characters

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Firefox prior to 2.0.0.12 had ignored 0x80 under Shift_JIS encoding. Firefox 2.0.0.12 以前のバージョンは
Shift_JIS のときに 0x80 を無視する Embedded invalid characters <s [0x80] c [0x80] r [0x80] ipt> alert(1)
</s [0x80] c [0x80] r [0x80] ipt>

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IE ignores 0x00. IE は 0x00 を 無視 する Embedded invalid characters <s [0x00] c [0x00] r [0x00] ipt>
alert(1) </s [0x00] c [0x00] r [0x00] ipt>

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IE considers 0x0B and 0x0C as delimiter. IE は 0x0B と 0x0C を 区切り文字 とみなす Embedded invalid characters <script [0x0B]

alert(1) </s cr ipt> <input type=text value= a [0x0C] onmouseover=alert(1)

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Countermeasure: Generate only safe string with white listing. ホワイトリストを用いて安全な文字列のみ生成する。 Embedded invalid characters

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Inject leading byte of Multi Byte Character Set(MBCS) to bypass filters マルチバイト文字の先行バイトを注入することでフィルタを回避 Embedded leading bytes

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Invalidate quotation with 0x82, leading byte of Shift_JIS. Shift_JIS の先行バイトである 0x82 でダブルクォートを無効にする Embedded leading bytes name: <input type=text value=" [0x82]"> e-mail: <input type=text value= " onmouseover=... //">

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Bypass XSS Filter of IE8 using leadbyte of MBCS. IE8 の XSS Filter も回避 Embedded leading bytes UTF-8
http://example.com/?%3cscript%20 %E2%3E alert(1);... http://example.com/? %E2%22
onmouseover=alert(1) Shift_JIS http://example.com/?%3cscript%20 %81%3E %3ealert(1);... EUC-JP

http://example.com/?%3cscript%20 %E0%3E alert(1);... http://example.com/? %E0%22
onmouseover=alert(1)

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Countermeasure: Validate by a letter unit. Convert another encoding... 文字単位で検証 他の文字コード
に変換 ... Embedded leading bytes

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Different understanding about the charset between server and client サーバとクライアント間で charset
の解釈が異なる Mismatch in charset information <html> < > Process Escape Generate HTML User
1101100110010 0010001110110 1000010100110 0101011011110 < → <

→ > " → " & → & ' → ' UTF-8 UTF-7

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Typical issue is XSS with UTF-7 When charset is ambiguous, IE assumes it as UTF-7 and causes XSS. 典型
的には UTF-7 による XSS が該当 charset が不明瞭なとき、IE は UTF-7 だと解釈して XSS が発生
Mismatch in charset information

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No charset is specified neither HTTP response header nor <meta> charset が指定されていない
Mismatch in charset information HTTP/1.1 200 OK Content-Type: text/html ... <html><head> <meta
http-equiv="content-type" content=" text/html "> </head><body> +ADw-script+AD4- alert(1)
+ADw-/script+AD4- ...

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Unrecognizable charset name for IE IE が解釈できない charset 名 Typically wrong charset names are:
CP932 / MS932 / sjis / jis / utf8 ... Mismatch in charset information <meta http-equiv='content-type'

content='text/html; charset=CP932 '> +ADw-script+AD4- alert(document.cookie); +ADw-/script+AD4-

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Unrecognizable charset name for IE Google, Yahoo, IBM ... IE doesn't recognize "CP932", "CP950", "EUC" for charset name Mismatch in charset information [http://www.google.com/search?oe=CP932&q=%2bADw- ... http://www.google.com/search?oe=CP950&q=%2bADw- ... http://search.yahoo.com/search?eo=EUC&p=%2bADw- ...](http://www.google.com/search?oe=CP932&q=%2bADw-...http://www.google.com/search?oe=CP950&q=%2bADw-...http://search.yahoo.com/search?eo=EUC&p=%2bADw-...)

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Inject fake <meta> before original it. 本来の <meta> より前に偽の <meta> を注入 Mismatch in charset information <title> +ADw-/title+AD4- +ADw-meta http-equiv+AD0-'content-type' content+AD0-'text/html+ADs-charset+AD0-utf-7'+AD4- </title> <meta http-equiv='content-type' content='text/html; charset= euc-jp '>

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Combination of UTF-7 with Ignoring Content-Type of IE. IE6 doesn't support "application/atom+xml" for Content-Type. Determine as UTF-7 HTML by content. Mismatch in charset information HTTP/1.1 200 OK Content-Type : application/atom+xml <?xml version='1.0' encoding= 'utf-8 '?>... <title>Search: +ADw-/title+AD4- +ADw-script+AD4- ... No charset

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Countermeasure for UTF-7 XSS : Specify charset clearly at HTTP response header. Specify recognizable charset name by browser. Don't place the text attacker can control before "<meta>". charset を HTTP レスポンスヘッダで明記する ブラウザが理解できる charset 名とする <meta> より前に攻撃者がコントロールできる文字列を置かない Mismatch in charset information

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UTF-7 issues affect not only IE and XSS, but also other browsers. UTF-7 の問題は IE での XSS だけでなく他のブラウザにも影響 Mismatch in charset information

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Yet Another JSON Hijacking with UTF-7 If no charset is specified in HTTP response header If attacker can control a part of JSON string Attacker can handle inside data of the JSON UTF-7 を使った JSON Hijacking HTTP レスポンスヘッダに charset が無い 攻撃者が JSON の一部をコントロール可能 JSON 内のデータを操作可能 Mismatch in charset information

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JSON Hijacking with UTF-7 Mismatch in charset information [{ "name" : " abc+MPv/fwAiAH0AXQA7-var t+AD0AWwB7ACIAIg-: +ACI- ", "mail" : "hasegawa@utf-8.jp" }, { "name" : "Kanatoko", "mail" : "anvil@example.com" }] JSON for target: http://example.com/target.json Injected by the attacker No charset in HTTP response header This means...

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JSON Hijacking with UTF-7 Mismatch in charset information [{ "name" : " abc"}];var t=[{"": " ", "mail" : "hasegawa@utf-8.jp" }, { "name" : "Kanatoko", "mail" : "anvil@example.com" }] JSON for target: http://example.com/target.json No charset in HTTP response header

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JSON Hijacking with UTF-7 Mismatch in charset information <script src="http://example.com/target.json" charset=" utf-7 "></script> <script> alert(t[1].name + t[1].mail); </script> Trap page: [{ "name" : " abc"}];var t=[{"": " ", "mail" : "hasegawa@utf-8.jp" }, { "name" : "Kanatoko", "mail" : "anvil@example.com" }] 外から JSON が UTF-7 であると指定。 setter が使えない場面でも有効。 Specify charset as UTF-7 from outside of JSON. No need to use **defineSetter**

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Mismatch in charset information DEMO

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Countermeasure for JSON: Place "while (1);" before JSON text. Accept only "POST", Reject access by "GET". while(1); を JSON の前に配置 POST のみ受け入れる Mismatch in charset information

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IE ignores the most significant bit of US-ASCII. IE は US-ASCII の 最上位ビットを無視する Interpreting 7-bit encoding 0010 0010 2 2 " 0x22 1 010 0010 A 2 「 0xA2 0011 1100 3 C < 0x3C 1 011 1100 B C シ 0xBC 0011 1110 3 E

0x3E セ 0xBE 1 011 1110 B E

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Interpreting 7-bit encoding

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MIME-Version: 1.0 Content-Type: text/plain; charset=US-ASCII Content-Transfer-Encoding: 7bit This is test mail begin 644 eicar.com ヲ #5/(5 E0\$%06S1<4%I8-30H4%X I-T-#*3=]) \$5)0T%2+5-404Y\$05)\$+4%.

75\$E625)54RU415-4+49)3\$4A)\$@K2"I# end OE also ignores the most significant bit of US-ASCII .

Interpreting 7-bit encoding ^ 0xCD M 0x4D カ 0xB6 6 0x36

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Countermeasure: Specify charset clearly on HTTP response header. Don't use US-ASCII. Use ISO-8859-1 and so on. HTTP レスポンスヘッダで charset を明記する US-ASCII を避け、ISO-8859-1 などを使う

Interpreting 7-bit encoding

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Deceptive indications 表示上の欺瞞

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Visual effect for human being Provoke a mistake Effective and useful tool for attackers 人間に対する視覚的な効果 ミスを誘う 攻撃者の強力な便利な道具 Deceptive indications

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Agenda はじめに 比較の一致 / 不一致 冗長なエンコーディング 多対一の変換 大文字と小文字 正規化 不正なバイト列の埋め込み 先行バイトの埋め込み エンコード情報の不一致 7 ビット文字コードの解釈 表示上の欺瞞 視覚的に似た文字 見えない文字 制御文字の埋め込み まとめ Introduction Comparison: match/unmatch Redundant encoding Many-to-one Conversion Upper case and Lower case

Normalization Embedded invalid characters Embedded leading bytes Mismatch in charset information
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characters Embedded control characters Conclusion

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Such as " 1 " (Digit One) and " l " (Small letter L)... <http://bank 1 .example.com/> <http://bank l .example.com/> More and more on Unicode ... 数字の 1 (イチ) と小文字の l (エル) など Unicode だともっとたくさん Characters with similar appearance

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Solidus and Division Slash Characters with similar appearance <http:// example.co.jp /t.example.com /foo/bar> Domain name / U+2215 / U+002F Solidus Division Slash

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Agenda はじめに 比較の一致 / 不一致 冗長なエンコーディング 多対一の変換 大文字と小文字 正規化
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Invisible byte sequence Unicode ISO-2022-JP Escape sequences Invisible characters U+200B ZERO
WIDTH SPACE U+200C ZERO WIDTH NON-JOINER U+200D ZERO WIDTH JOINER U+202A LEFT-TO-
RIGHT EMBEDDING U+FEFF BYTE ORDER MARK (ZWNBS) 0x1B 0x24 0x40 0x1B 0x24 0x42 0x1B 0x28
0x42

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Using for filename, registry Invisible characters

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Invisible characters DEMO

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Agenda はじめに 比較の一致 / 不一致 冗長なエンコーディング 多対一の変換 大文字と小文字 正規化
不正なバイト列の埋め込み 先行バイトの埋め込み エンコード情報の不一致 7 ビット文字コードの解釈

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Unicode Bidirection (Bidi) Part of string is displayed from RIGHT to LEFT U+202E (Right-to-Left Override;RLO) Unicode の双方向機能 (Bidi) 文字列の一部が右から左に表示される Embedded control characters this- (U+202E) txt.exe this-exe.txt Actual byte sequence Displayed text

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Embedded control characters this- (U+202E) txt.exe this-exe.txt Actual byte sequence Displayed text

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Embedded control characters DEMO

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Countermeasure: Prepare multiple confirmation methods SSL / EVSSL Display as Punycode Deceptive indications

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Conclusion まとめ

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Never convert to another encoding or normalize after validating strings. Don't be deceived only by an appearance. Security issues concerning character encodings are uncultivated fields. 検査後は変換・正規化しない 見た目だけに騙されない 文字コード × セキュリティって未開拓 Conclusion

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Yosuke HASEGAWA hasegawa@netagent.co.jp hasegawa@utf-8.jp http://utf-8.jp/ Questions?

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